

HHDMS

Home Healthcare & Diagnostic Center Management System

"Home Healthcare, One Call Away"

CSE 299 — North South University

Team: Name 1 Name 2 Name 3 Name 4

August 2026

The Problem & Our Solution

The Problem

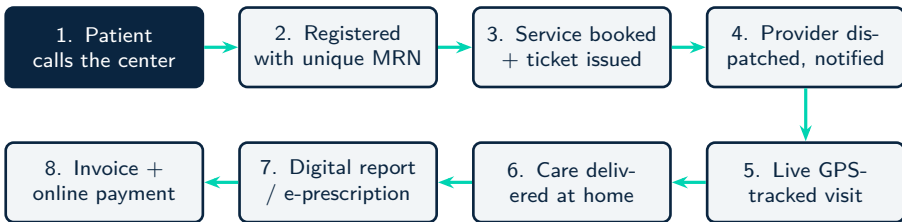
- Home healthcare in Bangladesh runs on phone calls — no tracking, no records, no accountability
- Families cannot see where the doctor or nurse is
- Prescriptions and reports are paper-only, easily lost
- Payments are informal; billing is opaque

Our Solution: One Platform

- A single call center books any service — doctor visit, nursing, diagnostics, nutrition
- Patients watch their provider arrive in real time on a map
- Digital records: e-prescriptions, reports, invoices
- Secure online payments and instant notifications

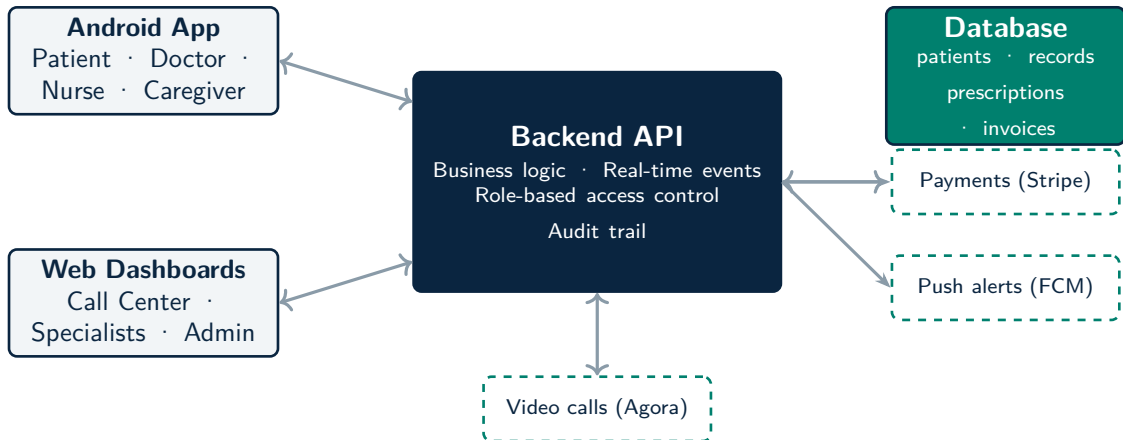
From an inbound phone call to a delivered home visit — fully digital.

How It Works — The Core Journey



Every step is timestamped and visible to the right people —
patient, provider, and operations staff.

System at a Glance



One Platform, Nine Roles

Call Center
intake & booking

Patient
app: track, chat, pay

MBBS Doctor
first clinical
assessment

Specialist
consults + DI-
COM imaging

Nurse
vitals, meds,
wound care

Caregiver
daily care,
GPS shifts

Nutritionist
diet plans
(EN + BN)

Sonologist
ultrasound studies

Admin Dashboard — live KPIs, staff management, financial oversight, tamper-proof audit logs

Demo I — Clinical Workflow

Screenshot
(paste image here)

**Call Center: patient registration & service
booking**

Screenshot
(paste image here)

Doctor: vitals → diagnosis → e-prescription

Script: register patient in under a minute — assign doctor — record abnormal vitals — ICD-10 coded diagnosis — signed digital prescription as PDF.

Demo II — Patient Experience

Screenshot
(paste image here)

Patient app: live GPS tracking of the visiting doctor

Screenshot
(paste image here)

Specialist: annotated DICOM imaging viewer

Script: patient watches the doctor approach on the map — specialist opens the ultrasound study, measures and annotates — report shared back to the care chain.

Under the Hood

Layer	Technology
Mobile app	Kotlin, Jetpack Compose
Web dashboards	Next.js (React), Tailwind
Backend API	NestJS (TypeScript), Socket.IO
Database	PostgreSQL + Prisma ORM
Video calls	Agora RTC
Imaging	DICOM viewer (Cornerstone3D)
Payments	Stripe
Notifications	Firebase Cloud Messaging

Scale of the build

- 69 database models
- ~100 REST endpoints
- 36 Android screens across 3 apps
- Real-time: chat, call signaling, live visit state

Built with Security in Mind

Medical-grade safeguards

- **Authenticated access** — every request carries a verified token (JWT)
- **Role-based access control** — each role sees only what it must
- **Tamper-proof audit trail** — admin actions recorded in a hash-chained log; any alteration is mathematically detectable
- **Encrypted transport** — HTTPS in production
- **Imaging protection** — DICOM studies served only through the authenticated viewer

Patient data is treated as the asset it is.

Where We Are — Status & Roadmap

Working today

- Full clinical workflows: MBBS → Specialist referrals
- Nursing, caregiver & nutritionist modules
- E-prescriptions with digital signatures
- Online payments, invoices & receipts
- Admin dashboard with live KPIs
- Push notifications & in-app chat

Next sprint (roadmap)

- X-Ray service module (booking → radiologist report)
- Fleet-wide live GPS dispatch engine
- Automated scheduling: reminders, rescheduling, recurring visits
- Web patient portal & feedback loop
- SMS / WhatsApp notification channels

Architecture is in place to absorb each addition without rework.

Healthcare that comes home.

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Questions?